

Solutions for Week 10 Maths Exercise Sheet

Vector Spaces and Matrices

Note that you will be given **feedback** for question (2).

(1) Throughout this question, the field we consider is the set of rational numbers.

- (a) Show that the two vectors $\begin{pmatrix} 1 \\ -4 \end{pmatrix}$ and $\begin{pmatrix} 0 \\ 0 \end{pmatrix}$ are not linearly independent, i.e., are linearly dependent, in the vector space $\mathbb{Q}^2$ of two-tuples of rational numbers.
- (b) Show that the four vectors $\begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}$, $\begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix}$, $\begin{pmatrix} 14 \\ 5 \\ \frac{1}{2} \end{pmatrix}$ and $\begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix}$ do not form a basis for the vector space $\mathbb{Q}^3$ of three-tuples of rational numbers. Show also that these four vectors are a spanning set.

Comment: Recall the definitions of linear independence and spanning set.

Solution:

- (a) Consider the linear combination where we multiply $\begin{pmatrix} 1 \\ -4 \end{pmatrix}$ by 0 and $\begin{pmatrix} 0 \\ 0 \end{pmatrix}$ by any non-zero constant, say 11. Then we have

$$0 \begin{pmatrix} 1 \\ -4 \end{pmatrix} \oplus 11 \begin{pmatrix} 0 \\ 0 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix} \oplus \begin{pmatrix} 0 \\ 0 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix} = \vec{0}$$

Therefore, there is a linear combination of $\begin{pmatrix} 1 \\ -4 \end{pmatrix}$ and $\begin{pmatrix} 0 \\ 0 \end{pmatrix}$ which is the zero vector $\begin{pmatrix} 0 \\ 0 \end{pmatrix}$ but one of the scalars is non-zero (here it was 11). Hence, the two vectors $\begin{pmatrix} 1 \\ -4 \end{pmatrix}$ and $\begin{pmatrix} 0 \\ 0 \end{pmatrix}$ are not linearly independent.

- (b) Recall that a set of vectors is a basis if it is **both** linearly independent and a spanning set. We show that the given four vectors are not linearly independent, and hence not a basis.

Three of these vectors $\begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}$, $\begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix}$, $\begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix}$ form the standard basis and hence we can easily observe that

$$\begin{pmatrix} 14 \\ 5 \\ \frac{1}{2} \end{pmatrix} = 14 \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix} \oplus 5 \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix} \oplus \frac{1}{2} \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix}$$

Rearranging this, we get

$$\vec{0} = \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix} = 14 \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix} \oplus 5 \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix} \oplus \frac{1}{2} \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix} \oplus (-1) \begin{pmatrix} 14 \\ 5 \\ \frac{1}{2} \end{pmatrix}$$

Therefore, we can express $\vec{0}$ as a linear combination of the four given vectors where none of the scalars are zero (note that even one of the four scalars being non-zero would have been sufficient for showing linear dependence!) since they are 14, 5, $\frac{1}{2}$ and (-1) .

To show that the four given vectors are a spanning set, we observe that three of the given vectors form the standard basis and hence also a spanning set. In particular, given any

vector $\begin{pmatrix} a \\ b \\ c \end{pmatrix} \in \mathbb{Q}^3$ we can show that it belongs to the span of the three vectors forming the standard basis as follows:

$$\begin{pmatrix} a \\ b \\ c \end{pmatrix} = a \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix} \oplus b \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix} \oplus c \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix}$$

Adding an extra vector (in this case $\begin{pmatrix} 14 \\ 5 \\ \frac{1}{2} \end{pmatrix}$) to a spanning set maintains the property that we still have a spanning set since we can just choose to multiply this new vector with the scalar 0. In other words, we can rewrite the above equation as follows:

$$\begin{pmatrix} a \\ b \\ c \end{pmatrix} = a \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix} \oplus b \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix} \oplus c \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix} \oplus 0 \begin{pmatrix} 14 \\ 5 \\ \frac{1}{2} \end{pmatrix}$$

Since we had chosen a general vector $\begin{pmatrix} a \\ b \\ c \end{pmatrix} \in \mathbb{Q}^3$ and shown that the given four vectors can span it, it follows that the given four vectors can span every vector in $\mathbb{Q}^3$, i.e., they form a spanning set for $\mathbb{Q}^3$.

(2) **(feedback)** Throughout this question, the field we consider is the set of rational numbers.

- (a) Show that the two vectors $\begin{pmatrix} 2 \\ -2 \end{pmatrix}$ and $\begin{pmatrix} 1 \\ 3 \end{pmatrix}$ form a basis for the vector space $\mathbb{Q}^2$ of two-tuples of rational numbers. Is this basis orthogonal?
- (b) Show that the three vectors $\begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}$, $\begin{pmatrix} 0 \\ 2 \\ 3 \end{pmatrix}$ and $\begin{pmatrix} 1 \\ 2 \\ 4 \end{pmatrix}$ form a basis for the vector space $\mathbb{Q}^3$ of three-tuples of rational numbers. Is this basis orthogonal?

Comment: Recall that for a set of vectors to be a basis you need to show two conditions: they are linearly independent, and they are a spanning set, i.e., every vector can be spanned by them. For a basis to be orthogonal, you need to check that inner product of every pair of distinct vectors in the basis is 0.

Solution:

- (a) First we show that the two vectors $\begin{pmatrix} 2 \\ -2 \end{pmatrix}$ and $\begin{pmatrix} 1 \\ 3 \end{pmatrix}$ are linearly independent. Suppose that there exists $x, y \in \mathbb{Q}$ such that $x \begin{pmatrix} 2 \\ -2 \end{pmatrix} \oplus y \begin{pmatrix} 1 \\ 3 \end{pmatrix} = \vec{0} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}$. From the first coordinate, i.e., the first row we get

$$2x + y = 0 \tag{1}$$

From the second coordinate, i.e., the second row we get

$$-2x + 3y = 0 \tag{2}$$

Adding equation (1) and equation (2), we get $4y = 0$ which implies $y = 0$. Substituting in equation (1) we get $2x = 0$ which implies $x = 0$. Hence, $x \begin{pmatrix} 2 \\ -2 \end{pmatrix} \oplus y \begin{pmatrix} 1 \\ 3 \end{pmatrix} = \vec{0} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}$ implies $x = 0 = y$. That is, the two vectors $\begin{pmatrix} 2 \\ -2 \end{pmatrix}$ and $\begin{pmatrix} 1 \\ 3 \end{pmatrix}$ are linearly independent.

We now show that these vectors form a spanning set for $\mathbb{Q}^2$ over the field $\mathbb{Q}$. That is, we need to show that every vector in $\mathbb{Q}^2$ belongs to the span of $\begin{pmatrix} 2 \\ -2 \end{pmatrix}$ and $\begin{pmatrix} 1 \\ 3 \end{pmatrix}$. Consider any general vector in $\mathbb{Q}^2$: how does it look like? It must be of the form $\begin{pmatrix} a \\ b \end{pmatrix}$ for some $a, b \in \mathbb{Q}$. We now show that this general vector $\begin{pmatrix} a \\ b \end{pmatrix}$ belongs to the span of $\begin{pmatrix} 2 \\ -2 \end{pmatrix}$ and $\begin{pmatrix} 1 \\ 3 \end{pmatrix}$ thereby showing that each vector in $\mathbb{Q}^2$ can be spanned by the given two vectors. We do this by solving for r_1, r_2 from the equation

$$r_1 \begin{pmatrix} 2 \\ -2 \end{pmatrix} \oplus r_2 \begin{pmatrix} 1 \\ 3 \end{pmatrix} = \begin{pmatrix} a \\ b \end{pmatrix} \quad (3)$$

From the first coordinate, i.e., the first row we get

$$2r_1 + r_2 = a \quad (4)$$

From the second coordinate, i.e., the second row we get

$$-2r_1 + 3r_2 = b \quad (5)$$

We now want to express r_1, r_2 in terms of a, b . This can be done by adding equation (4) and equation (5) which gives $4r_2 = a + b$, i.e., $r_2 = \frac{a+b}{4}$. Substituting in equation (4) gives $2r_1 = a - r_2 = a - \frac{a+b}{4} = \frac{3a-b}{4}$ which implies $r_1 = \frac{3a-b}{8}$. We verify our solution is correct by checking that

$$\frac{3a-b}{8} \cdot \begin{pmatrix} 2 \\ -2 \end{pmatrix} \oplus \frac{a+b}{4} \cdot \begin{pmatrix} 1 \\ 3 \end{pmatrix} = \begin{pmatrix} \frac{3a-b}{4} + \frac{a+b}{4} \\ -\frac{3a-b}{4} + \frac{3(a+b)}{4} \end{pmatrix} = \begin{pmatrix} a \\ b \end{pmatrix}$$

Finally, we observe that the basis given by the two vectors $\begin{pmatrix} 2 \\ -2 \end{pmatrix}$ and $\begin{pmatrix} 1 \\ 3 \end{pmatrix}$ is **not orthogonal** since their inner product is not equal to 0: this is because the inner product of these two vectors is $\begin{pmatrix} 2 \\ -2 \end{pmatrix} \cdot \begin{pmatrix} 1 \\ 3 \end{pmatrix} = (2 \times 1) + (-2 \times 3) = 2 + (-6) = (-4)$.

- (b) First we show that the three vectors $\begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}$, $\begin{pmatrix} 0 \\ 2 \\ 3 \end{pmatrix}$ and $\begin{pmatrix} 1 \\ 2 \\ 4 \end{pmatrix}$ are linearly independent.

Suppose that there exists $x, y, z \in \mathbb{Q}$ such that $x \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix} \oplus y \begin{pmatrix} 0 \\ 2 \\ 3 \end{pmatrix} \oplus z \begin{pmatrix} 1 \\ 2 \\ 4 \end{pmatrix} = \vec{0} = \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix}$.

From the first coordinate, i.e., the first row we get

$$x + z = 0 \quad (6)$$

From the second coordinate, i.e., the second row we get

$$2y + 2z = 0 \quad (7)$$

From the third coordinate, i.e., the third row we get

$$3y + 4z = 0 \quad (8)$$

Two times equation (7) gives

$$4y + 4z = 0 \quad (9)$$

Equation (9) minus equation (8) gives $y = 0$. Substituting this value of y in equation (7) gives $z = 0$. Substituting this value of z in equation (6) gives $x = 0$.

Hence, $x \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix} \oplus y \begin{pmatrix} 0 \\ 2 \\ 3 \end{pmatrix} \oplus z \begin{pmatrix} 1 \\ 2 \\ 4 \end{pmatrix} = \vec{0} = \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix}$. implies $x = y = z = 0$. That is, the three vectors $\begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}$, $\begin{pmatrix} 0 \\ 2 \\ 3 \end{pmatrix}$ and $\begin{pmatrix} 1 \\ 2 \\ 4 \end{pmatrix}$ are linearly independent.

We now show that these three given vectors form a spanning set for $\mathbb{Q}^3$ over the field $\mathbb{Q}$.

That is, we need to show that every vector in $\mathbb{Q}^3$ belongs to the span of $\begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}$, $\begin{pmatrix} 0 \\ 2 \\ 3 \end{pmatrix}$

and $\begin{pmatrix} 1 \\ 2 \\ 4 \end{pmatrix}$. Consider any general vector in $\mathbb{Q}^3$: how does it look like? It must be of the

form $\begin{pmatrix} a \\ b \\ c \end{pmatrix}$ for some $a, b, c \in \mathbb{Q}$. We now show that this general vector $\begin{pmatrix} a \\ b \\ c \end{pmatrix}$ belongs

to the span of $\begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}$, $\begin{pmatrix} 0 \\ 2 \\ 3 \end{pmatrix}$ and $\begin{pmatrix} 1 \\ 2 \\ 4 \end{pmatrix}$ thereby showing that each vector in $\mathbb{Q}^3$ can be spanned by the given three vectors. We do this by solving for r_1, r_2, r_3 from the equation

$$r_1 \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix} \oplus r_2 \begin{pmatrix} 0 \\ 2 \\ 3 \end{pmatrix} \oplus r_3 \begin{pmatrix} 1 \\ 2 \\ 4 \end{pmatrix} = \begin{pmatrix} a \\ b \\ c \end{pmatrix} \quad (10)$$

From the first coordinate, i.e., the first row we get

$$r_1 + r_3 = a \quad (11)$$

From the second coordinate, i.e., the second row we get

$$2r_2 + 2r_3 = b \quad (12)$$

From the third coordinate, i.e., the third row we get

$$3r_2 + 4r_3 = c \quad (13)$$

We now want to express r_1, r_2, r_3 in terms of a, b, c . Two times equation (12) gives

$$4r_2 + 4r_3 = 2b \quad (14)$$

Equation (14) minus equation (13) gives

$$r_2 = (2b - c)$$

Substituting this value of r_2 into equation (12) gives $2r_3 = b - 2r_2 = b - 2(2b - c) = 2c - 3b$ which implies

$$r_3 = \frac{2c - 3b}{2}$$

Substituting this value of r_3 into equation (11) gives

$$r_1 = a - r_3 = a - \frac{2c - 3b}{2} = \frac{2a + 3b - 2c}{2}$$

We verify our solution is correct by checking that

$$\begin{aligned} & \frac{2a+3b-2c}{2} \cdot \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix} \oplus (2b-c) \cdot \begin{pmatrix} 0 \\ 2 \\ 3 \end{pmatrix} \oplus \frac{2c-3b}{2} \cdot \begin{pmatrix} 1 \\ 2 \\ 4 \end{pmatrix} \\ &= \begin{pmatrix} \frac{2a+3b-2c}{2} + \frac{2c-3b}{2} \\ 2(2b-c) + (2c-3b) \\ 3(2b-c) + 2(2c-3b) \end{pmatrix} = \begin{pmatrix} a \\ b \\ c \end{pmatrix} \end{aligned}$$

Hence, the three vectors $\begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}$, $\begin{pmatrix} 0 \\ 2 \\ 3 \end{pmatrix}$ and $\begin{pmatrix} 1 \\ 2 \\ 4 \end{pmatrix}$ form a basis. For this basis to be

orthogonal, we need **each of the three** inner products $\begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix} \cdot \begin{pmatrix} 0 \\ 2 \\ 3 \end{pmatrix}$, $\begin{pmatrix} 0 \\ 2 \\ 3 \end{pmatrix} \cdot \begin{pmatrix} 1 \\ 2 \\ 4 \end{pmatrix}$ and

$\begin{pmatrix} 1 \\ 2 \\ 4 \end{pmatrix} \cdot \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}$ to be equal to 0. Hence, this basis is not orthogonal as

$$\begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix} \cdot \begin{pmatrix} 1 \\ 2 \\ 4 \end{pmatrix} = (1 \times 1) + (0 \times 2) + (0 \times 4) = 1$$

Note that here one of the inner products viz. $\begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix} \cdot \begin{pmatrix} 0 \\ 2 \\ 3 \end{pmatrix} = 0$ but this is not sufficient for the basis of the three given vectors to be orthogonal.

(3) This question is about matrix multiplication. Let $A = \begin{pmatrix} 1 & 5 & 8 \\ 0 & 6 & 7 \end{pmatrix}$ and $B = \begin{pmatrix} 18 & 7 \\ 11 & -4 \end{pmatrix}$.

- (a) Can you multiply A with B to obtain the matrix AB ? If yes, find the matrix AB and show your steps? If not, explain why.
- (b) Can you multiply B with A to obtain the matrix BA ? If yes, find the matrix BA and show your steps? If not, explain why.

Solution

To multiply two matrices X and Y to obtain the matrix XY , we need number of columns of the first X to be equal to number of rows of the second matrix Y .

A is a 2×3 matrix, i.e, has 2 rows and 3 columns. B is a 2×2 matrix, i.e, has 2 rows and 2 columns. Hence, AB is not defined here but BA is defined. We now compute the matrix BA using the definition of matrix multiplication:

$$\begin{aligned} BA &= \begin{pmatrix} 18 & 7 \\ 11 & -4 \end{pmatrix} \times \begin{pmatrix} 1 & 5 & 8 \\ 0 & 6 & 7 \end{pmatrix} = \begin{pmatrix} (18 \times 1) + (7 \times 0) & (18 \times 5) + (7 \times 6) & (18 \times 8) + (7 \times 7) \\ (11 \times 1) + (-4 \times 0) & (11 \times 5) + (-4 \times 6) & (11 \times 8) + (-4 \times 7) \end{pmatrix} \\ &= \begin{pmatrix} 18 & 132 & 193 \\ 11 & 31 & 60 \end{pmatrix} \end{aligned}$$

- (4) Let $a, b, c, d \in \mathbb{Q}$ be such that $(ad-bc) \neq 0$. Let $X = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$ and $Y = \begin{pmatrix} \frac{d}{ad-bc} & \frac{-b}{ad-bc} \\ \frac{-c}{ad-bc} & \frac{a}{ad-bc} \end{pmatrix}$.

Show that $XY = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} = YX$.

Comment: Note that you are asked to compute two matrix multiplications in this question.

Solution: First we compute XY as follows:

$$XY = \begin{pmatrix} a \cdot \frac{d}{ad-bc} + b \cdot \frac{-c}{ad-bc} & a \cdot \frac{-b}{ad-bc} + b \cdot \frac{a}{ad-bc} \\ c \cdot \frac{d}{ad-bc} + d \cdot \frac{-c}{ad-bc} & c \cdot \frac{-b}{ad-bc} + d \cdot \frac{a}{ad-bc} \end{pmatrix} = \begin{pmatrix} \frac{ad-bc}{ad-bc} & \frac{0}{ad-bc} \\ \frac{0}{ad-bc} & \frac{ad-bc}{ad-bc} \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$$

Similarly, we can compute YX

$$YX = \begin{pmatrix} \frac{d}{ad-bc} \cdot a + \frac{-b}{ad-bc} \cdot c & \frac{d}{ad-bc} \cdot b + \frac{-b}{ad-bc} \cdot d \\ \frac{-c}{ad-bc} \cdot a + \frac{a}{ad-bc} \cdot c & \frac{-c}{ad-bc} \cdot b + \frac{a}{ad-bc} \cdot d \end{pmatrix} = \begin{pmatrix} \frac{ad-bc}{ad-bc} & \frac{0}{ad-bc} \\ \frac{0}{ad-bc} & \frac{ad-bc}{ad-bc} \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$$

- (5) Verify that the set of all (2×2) matrices $\left\{ \begin{pmatrix} a & b \\ c & d \end{pmatrix} \mid a, b, c, d \in \mathbb{Q} \right\}$ is a vector space over the field $\mathbb{Q}$ (which has the two operations $+$ and $\times$) with the following definition of vector addition & multiplication of a scalar with a vector:

- (a) Vector addition: If $\vec{v}_1 = \begin{pmatrix} a_1 & b_1 \\ c_1 & d_1 \end{pmatrix}$ and $\vec{v}_2 = \begin{pmatrix} a_2 & b_2 \\ c_2 & d_2 \end{pmatrix}$ then $\vec{v}_1 \oplus \vec{v}_2$ is defined to be
- $$\begin{pmatrix} a_1 + a_2 & b_1 + b_2 \\ c_1 + c_2 & d_1 + d_2 \end{pmatrix}$$

- (b) Multiplication of a scalar and a vector: If $\vec{v} = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$ and $r \in \mathbb{Q}$ then $r\vec{v} = \begin{pmatrix} r \times a & r \times b \\ r \times c & r \times d \end{pmatrix}$

Comment: Each of the 8 conditions in the definition of vector space over a field needs to be verified. You can use any of the conditions available due to $\mathbb{Q}$ being a field.

Remark : The addition and multiplication of elements within $\mathbb{Q}$ is denoted by $+$ and $\times$ respectively. The addition of two vectors from the set of all (2×2) matrices $\left\{ \begin{pmatrix} a & b \\ c & d \end{pmatrix} \mid a, b, c, d \in \mathbb{Q} \right\}$ is via the addition operation denoted by $\oplus$. The multiplication of a scalar s and a vector $\vec{v}$ is denoted by $s\vec{v}$.

Solution:

Recall that $\mathbb{Q}$ being a field gives us the following properties we can use for solving this question:

$$a + 0 = a \qquad a \times 1 = a \qquad (\text{neutral elements})$$

$$a + b = b + a \qquad a \times b = b \times a \qquad (\text{commutativity})$$

$$(a + b) + c = a + (b + c) \qquad (a \times b) \times c = a \times (b \times c) \quad (\text{associativity})$$

$$a + (-a) = 0 \qquad a \times a^{-1} = 1 \quad (a \neq 0) \quad (\text{inverses})$$

$$a \times (b + c) = a \times b + a \times c \qquad (\text{distributivity of } \times \text{ over } +)$$

Having defined vector addition & multiplication of a scalar and a vector, we are now ready to verify the 8 conditions for $\mathbb{Q}^2$ to be a vector space over the field $\mathbb{Q}$ of rational numbers. Let $\vec{u} = \begin{pmatrix} a_1 & b_1 \\ c_1 & d_1 \end{pmatrix}$, $\vec{v} = \begin{pmatrix} a_2 & b_2 \\ c_2 & d_2 \end{pmatrix}$ and $\vec{w} = \begin{pmatrix} a_3 & b_3 \\ c_3 & d_3 \end{pmatrix}$ for some $a, b, c, d, e, f \in \mathbb{Q}$. Let $r, s \in \mathbb{Q}$ be scalars.

(1) Commutativity of vector addition:

$$\begin{aligned} \vec{u} \oplus \vec{v} &= \begin{pmatrix} a_1 & b_1 \\ c_1 & d_1 \end{pmatrix} \oplus \begin{pmatrix} a_2 & b_2 \\ c_2 & d_2 \end{pmatrix} \\ &= \begin{pmatrix} a_1 + a_2 & b_1 + b_2 \\ c_1 + c_2 & d_1 + d_2 \end{pmatrix} && (\text{by definition of } \oplus) \\ &= \begin{pmatrix} a_2 + a_1 & b_2 + b_1 \\ c_2 + c_1 & d_2 + d_1 \end{pmatrix} && (\text{by commutativity of } + \text{ in } \mathbb{Q}) \\ &= \begin{pmatrix} a_2 & b_2 \\ c_2 & d_2 \end{pmatrix} \oplus \begin{pmatrix} a_1 & b_1 \\ c_1 & d_1 \end{pmatrix} && (\text{by definition of } \oplus) \\ &= \vec{v} \oplus \vec{u} \end{aligned}$$

(2) Associativity of vector addition:

$$\begin{aligned} (\vec{u} \oplus \vec{v}) \oplus \vec{w} &= \left(\begin{pmatrix} a_1 & b_1 \\ c_1 & d_1 \end{pmatrix} \oplus \begin{pmatrix} a_2 & b_2 \\ c_2 & d_2 \end{pmatrix} \right) \oplus \begin{pmatrix} a_3 & b_3 \\ c_3 & d_3 \end{pmatrix} \\ &= \begin{pmatrix} (a_1 + a_2) & (b_1 + b_2) \\ (c_1 + c_2) & (d_1 + d_2) \end{pmatrix} \oplus \begin{pmatrix} a_3 & b_3 \\ c_3 & d_3 \end{pmatrix} && (\text{by definition of } \oplus) \\ &= \begin{pmatrix} (a_1 + a_2) + a_3 & (b_1 + b_2) + b_3 \\ (c_1 + c_2) + c_3 & (d_1 + d_2) + d_3 \end{pmatrix} && (\text{by definition of } \oplus) \\ &= \begin{pmatrix} a_1 + (a_2 + a_3) & b_1 + (b_2 + b_3) \\ c_1 + (c_2 + c_3) & d_1 + (d_2 + d_3) \end{pmatrix} && (\text{by associativity of } + \text{ in } \mathbb{Q}) \\ &= \begin{pmatrix} a_1 & b_1 \\ c_1 & d_1 \end{pmatrix} \oplus \begin{pmatrix} (a_2 + a_3) & (b_2 + b_3) \\ (c_2 + c_3) & (d_2 + d_3) \end{pmatrix} && (\text{by definition of } \oplus) \\ &= \begin{pmatrix} a_1 & b_1 \\ c_1 & d_1 \end{pmatrix} \oplus \left(\begin{pmatrix} a_2 & b_2 \\ c_2 & d_2 \end{pmatrix} \oplus \begin{pmatrix} a_3 & b_3 \\ c_3 & d_3 \end{pmatrix} \right) && (\text{by definition of } \oplus) \\ &= \vec{u} \oplus (\vec{v} \oplus \vec{w}) \end{aligned}$$

- (3) Existence of Additive identity: We claim that $\vec{0} = \begin{pmatrix} 0 & 0 \\ 0 & 0 \end{pmatrix}$ is an additive identity for this vector space. Consider any vector $\vec{u} = \begin{pmatrix} a_1 & b_1 \\ c_1 & d_1 \end{pmatrix}$ which is a (2×2) matrix with rational entries. Then we have

$$\begin{aligned} \vec{0} \oplus \vec{u} &= \begin{pmatrix} 0 & 0 \\ 0 & 0 \end{pmatrix} \oplus \begin{pmatrix} a_1 & b_1 \\ c_1 & d_1 \end{pmatrix} \\ &= \begin{pmatrix} 0 + a_1 & 0 + b_1 \\ 0 + c_1 & 0 + d_1 \end{pmatrix} && \text{(by definition of } \oplus \text{)} \\ &= \begin{pmatrix} a_1 & b_1 \\ c_1 & d_1 \end{pmatrix} && \text{(since 0 is additive identity for } \mathbb{Q} \text{)} \\ &= \vec{u} \end{aligned}$$

- (4) Existence of additive inverse: Consider any vector $\vec{u} = \begin{pmatrix} a_1 & b_1 \\ c_1 & d_1 \end{pmatrix}$ which is a (2×2) matrix with rational entries. Since $a_1, b_1, c_1, d_1 \in \mathbb{Q}$ it follows that they have additive inverses (with respect to the addition operation $+$ from the field $\mathbb{Q}$). That is, there exist $-a_1, -b_1, -c_1, -d_1 \in \mathbb{Q}$ such that

$$a_1 + (-a_1) = b_1 + (-b_1) = c_1 + (-c_1) = d_1 + (-d_1) = 0 \quad (15)$$

We claim that the vector $\begin{pmatrix} -a_1 & -b_1 \\ -c_1 & -d_1 \end{pmatrix}$ is an additive inverse for $\vec{u}$. This is because

$$\begin{aligned} \vec{u} \oplus \begin{pmatrix} -a_1 & -b_1 \\ -c_1 & -d_1 \end{pmatrix} &= \begin{pmatrix} a_1 & b_1 \\ c_1 & d_1 \end{pmatrix} \oplus \begin{pmatrix} -a_1 & -b_1 \\ -c_1 & -d_1 \end{pmatrix} \\ &= \begin{pmatrix} a_1 + (-a_1) & b_1 + (-b_1) \\ c_1 + (-c_1) & d_1 + (-d_1) \end{pmatrix} && \text{(by definition of } \oplus \text{)} \\ &= \begin{pmatrix} 0 & 0 \\ 0 & 0 \end{pmatrix} && \text{(by equation (15))} \\ &= \vec{0} \end{aligned}$$

- (5) Associativity of multiplication of scalar & vector: Consider any vector $\vec{u} = \begin{pmatrix} a_1 & b_1 \\ c_1 & d_1 \end{pmatrix}$ which is a (2×2) matrix with rational entries. Then we have

$$\begin{aligned} r(s\vec{u}) &= r \begin{pmatrix} s \times a_1 & s \times b_1 \\ s \times c_1 & s \times d_1 \end{pmatrix} && \text{(by definition of multiplication of a scalar and a vector)} \\ &= \begin{pmatrix} r \times (s \times a_1) & r \times (s \times b_1) \\ r \times (s \times c_1) & r \times (s \times d_1) \end{pmatrix} \\ &&& \text{(by definition of multiplication of a scalar and a vector)} \\ &= \begin{pmatrix} (r \times s) \times a_1 & (r \times s) \times b_1 \\ (r \times s) \times c_1 & (r \times s) \times d_1 \end{pmatrix} \\ &&& \text{(by associativity of multiplication operation } \times \text{ in } \mathbb{Q} \text{)} \\ &= (rs) \begin{pmatrix} a_1 & b_1 \\ c_1 & d_1 \end{pmatrix} && \text{(by definition of multiplication of a scalar and a vector)} \\ &= rs(\vec{u}) \end{aligned}$$

- (6) Distributivity of scalar sums: Consider any vector $\vec{u} = \begin{pmatrix} a_1 & b_1 \\ c_1 & d_1 \end{pmatrix}$ which is a (2×2) matrix with rational entries. Then we have

$$\begin{aligned}
 (r+s)\vec{v} &= (r+s) \begin{pmatrix} a_1 & b_1 \\ c_1 & d_1 \end{pmatrix} \\
 &= \begin{pmatrix} (r+s) \times a_1 & (r+s) \times b_1 \\ (r+s) \times c_1 & (r+s) \times d_1 \end{pmatrix} \\
 &\quad \text{(by definition of multiplication of a scalar and a vector)} \\
 &= \begin{pmatrix} r \times a_1 + s \times a_1 & r \times b_1 + s \times b_1 \\ r \times c_1 + s \times c_1 & r \times d_1 + s \times d_1 \end{pmatrix} \quad \text{(by distributivity of } \times \text{ over } + \text{ in } \mathbb{Q}) \\
 &= \begin{pmatrix} r \times a_1 & r \times b_1 \\ r \times c_1 & r \times d_1 \end{pmatrix} \oplus \begin{pmatrix} s \times a_1 & s \times b_1 \\ s \times c_1 & s \times d_1 \end{pmatrix} \quad \text{(by definition of } \oplus) \\
 &= r \begin{pmatrix} a_1 & b_1 \\ c_1 & d_1 \end{pmatrix} \oplus s \begin{pmatrix} a_1 & b_1 \\ c_1 & d_1 \end{pmatrix} \\
 &\quad \text{(by definition of multiplication of a scalar and a vector)} \\
 &= r\vec{v} \oplus s\vec{v}
 \end{aligned}$$

- (7) Distributivity of vector sums: Consider any two vector $\vec{u} = \begin{pmatrix} a_1 & b_1 \\ c_1 & d_1 \end{pmatrix}$ and $\vec{v} = \begin{pmatrix} a_2 & b_2 \\ c_2 & d_2 \end{pmatrix}$ which are (2×2) matrices with rational entries. Let $r \in \mathbb{Q}$. Then we have

$$\begin{aligned}
 r(\vec{u} \oplus \vec{v}) &= r \left(\begin{pmatrix} a_1 & b_1 \\ c_1 & d_1 \end{pmatrix} \oplus \begin{pmatrix} a_2 & b_2 \\ c_2 & d_2 \end{pmatrix} \right) \\
 &= r \begin{pmatrix} a_1 + a_2 & b_1 + b_2 \\ c_1 + c_2 & d_1 + d_2 \end{pmatrix} \quad \text{(by definition of } \oplus) \\
 &= \begin{pmatrix} r \times (a_1 + a_2) & r \times (b_1 + b_2) \\ r \times (c_1 + c_2) & r \times (d_1 + d_2) \end{pmatrix} \\
 &\quad \text{(by definition of multiplication of a scalar and a vector)} \\
 &= \begin{pmatrix} r \times a_1 + r \times a_2 & r \times b_1 + r \times b_2 \\ r \times c_1 + r \times c_2 & r \times d_1 + r \times d_2 \end{pmatrix} \quad \text{(by distributivity of } \times \text{ over } + \text{ in } \mathbb{Q}) \\
 &= \begin{pmatrix} r \times a_1 & r \times b_1 \\ r \times c_1 & r \times d_1 \end{pmatrix} \oplus \begin{pmatrix} r \times a_2 & r \times b_2 \\ r \times c_2 & r \times d_2 \end{pmatrix} \quad \text{(by definition of } \oplus) \\
 &= r \begin{pmatrix} a_1 & b_1 \\ c_1 & d_1 \end{pmatrix} \oplus r \begin{pmatrix} a_2 & b_2 \\ c_2 & d_2 \end{pmatrix} \\
 &\quad \text{(by definition of multiplication of a scalar and a vector)} \\
 &= r\vec{u} \oplus r\vec{v}
 \end{aligned}$$

- (8) Existence of identity of multiplication of scalar & vector: $1\vec{v} = \vec{v}$
 Since $\mathbb{Q}$ is a field we have a multiplicative identity (i.e., neutral element) 1 for multiplication of two scalars from $\mathbb{Q}$. That is, $1 \times q = q$ for each $q \in \mathbb{Q}$. We now claim that this same 1 is also a multiplicative identity for multiplication of a scalar and a vector. Consider any

vector $\vec{u} = \begin{pmatrix} a_1 & b_1 \\ c_1 & d_1 \end{pmatrix}$ which is a (2×2) matrix with rational entries. Then we have

$$\begin{aligned} 1\vec{u} &= 1 \begin{pmatrix} a_1 & b_1 \\ c_1 & d_1 \end{pmatrix} \\ &= \begin{pmatrix} 1 \times a_1 & 1 \times b_1 \\ 1 \times c_1 & 1 \times d_1 \end{pmatrix} && \text{(by definition of multiplication of a scalar and a vector)} \\ &= \begin{pmatrix} a_1 & b_1 \\ c_1 & d_1 \end{pmatrix} && \text{since 1 is identity for multiplication operation } \times \text{ in } \mathbb{Q} \\ &= \vec{u} \end{aligned}$$